Phishing Email Detection & Awareness Report

cyber security internship program

-FUTURE INTERNS

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01. Introduction & Project Objectives

Introduction:

Phishing is a type of cyber attack where scammers send fake emails to trick people into giving away sensitive information. These emails often look like they come from a trusted source, such as a bank, a social media platform, or even your own company's IT department. Instead of hacking into a computer system, attackers "hack" the person by using fear or urgency to make them click a malicious link or download a virus. Because these attacks target people rather than software, staying alert is our best line of defense.

Project Objectives:

The objective of this task is to provide a comprehensive analysis of a phishing threat to ensure all stakeholders can recognize and neutralize similar attacks. Throughout this report, the following phases will take place:

- **Analyze the Anatomy of an Attack:** examine a real-world phishing sample to expose the specific methods used to deceive recipients.
- **Identify Malicious Indicators:** pinpoint the "Red Flags," such as spoofed domains and deceptive hyperlinks, that reveal an email's true intent.
- **Classify Associated Risks:** evaluate the potential impact of the attack, ranging from credential theft to unauthorized network access.
- **Provide Awareness Guidelines:** deliver clear, actionable protocols to help users recognize, report, and prevent phishing attempts effectively.

02. Phishing Email Sample

The analyzed email is a high-fidelity phishing lure designed to impersonate an official Microsoft security notification. This specific sample was selected because it demonstrates a sophisticated attempt to bypass user suspicion through the use of established branding and psychological triggers.

Subject: Microsoft account unusual signin activity

(unknown sender)
to

Microsoft account

Unusual sign.in activity

We detected something unusual about a recent sign-in to the Microsoft account phishing@pot.

Sign-in details

Country/region: **Russia/Moscow**

IP address: **103.225.77.255**

Date: **Thu, 27 Jul 2023 07:40:03 +0000**

Platform: **Windows 10**

Browser: **Firefox**

A user from **Russia/Moscow** just logged into your account from a new device, If this wasn't you, please report the user. If this was you, we'll trust similar activity in the future.

Report The User

To opt out or change where you receive security notifications, click here.

Thanks,
The Microsoft account team

Analysis of the Phishing Lure

- **Urgency & Fear**: Uses a sign-in from **"Russia/Moscow"** to panic the user into acting quickly.
- **Fake Identity**: Impersonates the **"Microsoft account team"** to gain immediate trust.
- **Service Warning**: Gmail identifies this as an **"(unknown sender)"** because it cannot verify the source.
- **Malicious Button**: The **"Report The User"** button is a trap designed to steal credentials or start a fake support chat.
- **Generic Greeting**: The email is sent to a general address rather than a specific person's name.

03. Analysis of Email Headers

Google Admin Toolbox Messageheader Help

MessageId	412928a2-0aad-4b27-959a-5df404e1f07d@VI1EUR06FT066.eop-eur06.prod.protection.outlook.com
Created at:	7/27/2023, 1:10:03 PM GMT+5:30 (Delivered after 3 sec)
From:	Microsoft account team ,_<no-reply@access-accsecurity.com>
To:	phishing@pot
Subject:	Microsoft account unusual signin activity
SPF:	none Learn more

#	Delay	From *		To *	Protocol	Time received
0		nonkfrgr.co.uk	→	VI1EUR06FT066.mail.protection.outlook.com		7/27/2023, 1:10:03 PM GMT+5:30
1		VI1EUR06FT066.eop-eur06.prod.protection.outlook.com	→	AM6P194CA0105.outlook.office365.com		7/27/2023, 1:10:03 PM GMT+5:30
2	1 sec	AM6P194CA0105.EURP194.PROD.OUTLOOK.COM	→	PH7PR19MB5920.namprd19.prod.outlook.com		7/27/2023, 1:10:04 PM GMT+5:30
3	2 sec	PH7PR19MB5920.namprd19.prod.outlook.com	→	MN0PR19MB6312.namprd19.prod.outlook.com		7/27/2023, 1:10:06 PM GMT+5:30

The analyzed technical metadata reveals critical failures in the email's authentication and routing, confirming that the message is a sophisticated forgery.

- **Authentication Failure**: The header analysis shows an **SPF (Sender Policy Framework)** status of **"none"**. This indicates that the sender's IP address (89.144.9.87) is not authorized to send emails on behalf of the claimed domain.
- **Identity Discrepancy**: While the "From" field displays "Microsoft account team," the **Return-Path** is recorded as bounce@nonkfrgr.co.uk. This domain has no legitimate affiliation with Microsoft Corporation.
- **Routing Origin (MTA Hops)**: The Message header tool identifies the initial hop originating from nonkfrgr.co.uk. The routing path shows the email passing through multiple unauthorized servers before delivery.
- **Delivery Delay**: A **3-second delay** is observed between the initial dispatch and final delivery, which is a common characteristic of automated phishing scripts utilizing mail relays to mask their origin.

04. Authentication & Security Protocols:

Beyond the routing analysis, the email was audited against industry-standard security protocols to verify the sender's authenticity. The following evidence confirms a complete failure of all identity verification checks:

SPF and DKIM Information

dmarc:access-accsecurity.com Hide Solve Email Delivery Problems

	Test	Result	
✖	DMARC Record Published	No DMARC Record found	More Info

Reported by **k.gtld-servers.net** on 3/9/2026 at **4:21:26 AM (UTC 0)**, just for you. Transcript

spf:nonkfrgr.co.uk:::1 Show Solve Email Delivery Problems

```
Dkim Signature Error:
No DKIM-Signature header found - more info
```

```
Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned. - more info
```

Headers Found

Header Name	Header Value

DMARC Failure: An audit of the spoofed domain access-accsecurity.com confirms that **no DMARC record was found**. This critical lack of security policy allows attackers to impersonate the "Microsoft account team" without being blocked by automated mail filters.

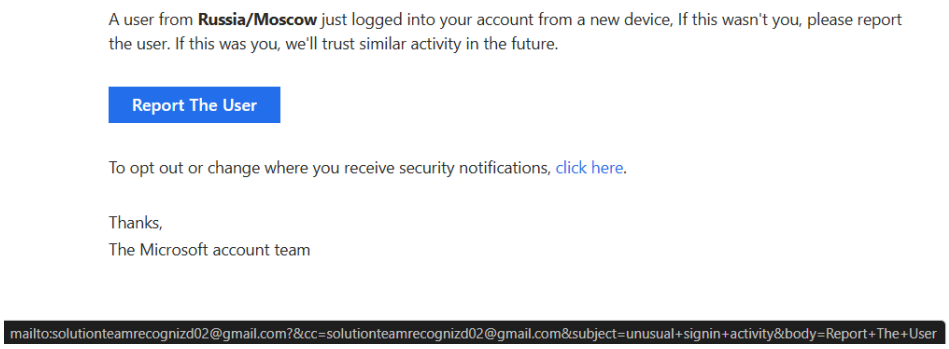
DKIM Authentication Error: The technical inspection reveals that **no DKIM-Signature header was found** in the message metadata. Because the email lacks a digital signature, its origin is unverified, and its contents could have been manipulated in transit.

SPF Mismatch: The **Sender Policy Framework (SPF)** status is explicitly marked as **"none"**. This proves that the sending IP address (89.144.9.87) is not authorized to dispatch emails on behalf of the claimed organization.

Protocol Summary: These three failures together confirm that the email is a "spoofed" message, using a fake "From" address to mask an unauthorized and malicious source.

05. Identification of Phishing Indicators:

The visual interface uses a sign-in alert from "Russia/Moscow" to create immediate panic, a psychological tactic intended to force a quick, unverified response from the user. While the sender name appears as "Microsoft account team," the actual address is <no-reply@access-accsecurity.com>, which has no official affiliation with Microsoft. This discrepancy between the display name and the hidden domain is a primary indicator of identity deception used to build false trust.



Technical inspection of the "Report The User" button reveals a "mailto" command rather than a link to a secure verification portal. As shown in the metadata, clicking this button initiates a direct email draft to a personal Gmail account (solutionteamrecognizd02@gmail.com) instead of a corporate security system. This is a definitive sign of credential harvesting, where attackers try to trick victims into sending sensitive information or codes under the guise of official support.

06. Classification of Email Risk Level:

Category	Analysis Detail
Risk Level	CRITICAL (High Priority)
Threat Type	Credential Harvesting / Identity Theft
Technical Status	SPF/DKIM/DMARC Fail (Spoofed Domain)
Target Asset	Microsoft Account Credentials & MFA Codes
Psychological Tactic	Fear & Urgency (Login from Russia/Moscow)
Potential Impact	Unauthorized Data Access & Account Takeover

07. Prevention and Awareness Guidelines

Focus Area	Actionable Guideline
Verify Sender	Always check the actual email address behind the display name; legitimate Microsoft alerts only come from @microsoft.com.
Inspect Links	Hover over buttons to preview the URL or "**mailto**" command before clicking; never interact with suspicious external domains.
Enable MFA	Use Multi-Factor Authentication (MFA) via an authenticator app to ensure account safety even if credentials are stolen.
Report Threats	Use the official "Report Phishing" button in your email client instead of clicking links within a suspicious message.
Avoid Panic	Be wary of emails using "Fear or Urgency" (like login alerts from foreign countries) to pressure you into immediate action.

08. Conclusion

The forensic analysis of the Microsoft "Unusual Sign-in" email confirms that even highly convincing lures contain technical and logical flaws. While the visual branding is sophisticated, the failure of authentication protocols and the presence of suspicious "mailto" links provide definitive proof of a phishing attempt.

This report demonstrates that a combination of technical auditing and user awareness is the most effective defense against credential harvesting. Utilizing tools for header analysis and identifying psychological triggers ensures that malicious threats are neutralized before they can compromise sensitive data.